

Statistical Evaluation of Protein and Sugar Nutritional Values

A hypothesis testing activity submitted by

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1. About the Dataset

The dataset used in this analysis contains nutritional values of protein and sugar, measured in grams (g). Each record represents the amount of protein and sugar associated with a specific calorie classification.

Source: <https://www.kaggle.com/datasets/amineipad/food-nutrition-dataset-for-calorie-classification>

2. Statistical Question

Is the nutritional value of protein significantly higher than that of sugar?

3. Hypotheses

$$H_0: \mu_{\text{Protein}} \leq \mu_{\text{Sugar}}$$

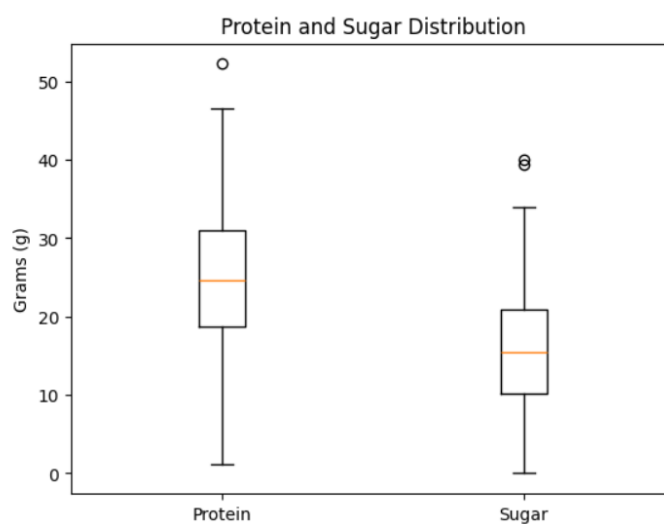
$$H_a: \mu_{\text{Protein}} > \mu_{\text{Sugar}}$$

4. Test Statistic

4.1 Summary of statistics

	protein_g	sugar_g
count	100.000000	100.000000
mean	24.758000	15.317000
std	9.832251	8.879906
min	1.100000	0.000000
25%	18.650000	10.150000
50%	24.650000	15.400000
75%	30.925000	20.850000
max	52.200000	40.000000

4.2 Boxplot



4.3 Normality Test

```
Protein:  
w_stat: 0.9934  
P-value: 0.9135  
Normal data
```

```
Sugar:  
w_stat: 0.9755  
P-value: 0.0588  
Normal data
```

4.4 F-Test

```
f_stat = 1.2260  
p_value = 0.1562  
equal variances
```

4.5 2-Sample t-Test

```
t_stat = 7.1260  
p_value = 0.0000  
Protein nutritional value is significantly higher than sugar nutritional value.
```

4.6 Power of the Test

```
Cohen's d: 0.9602  
Power of the test: 100.00%
```

5. Conclusion

Using a two-sample t-test to evaluate whether protein nutritional values are greater than sugar nutritional values, the analysis yielded a t-statistic of 51.9760 with a p-value < 0.0000. At the 0.05 significance level, the null hypothesis is rejected. This result indicates that the mean protein value is significantly higher than the mean sugar value in the dataset.